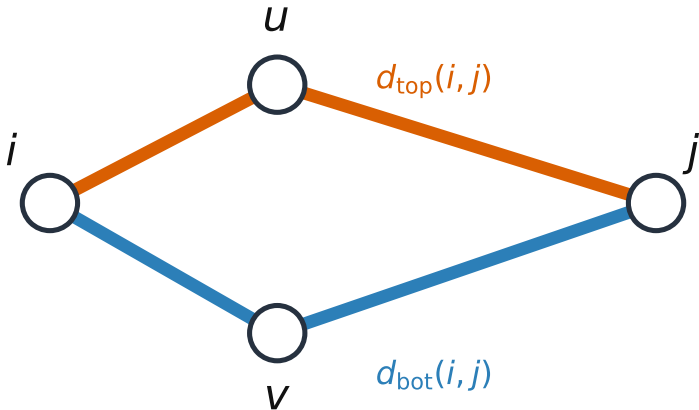


(b) Path switching / tie point

equal-length routes create a nonsmooth point in $d_k^{\text{emb}}(i, j; \mathbf{Z})$



$d_{\text{top}} = d_{\text{bot}} \Rightarrow \partial d_k^{\text{emb}}(i, j; \mathbf{Z})$ uses convex combinations